ZERTIFIKAT ◆ CERTIFICATE ◆ 認証書 ◆ СЕРТИФИКАТ ◆ CERTIFICADO ◆ CERTIFICAT

Product Service

EC-Attestation of Compliance

No. E8N 07 10 46532 011

Holder of Certificate: **AJINEXTEK CO., LTD.**
9-3, Holim-dong, dalseo-gu
Daegu-city 704-240
REPUBLIC OF KOREA

Name of Object: **Information Technology Equipment**
PCI One Motion Control Board

Model(s): **PCI-N804 and PCI-N404**

Description of Object:

Rated input voltage:	230 V a.c.
Rated input frequency:	50 Hz

Tested according to:

EN 55022:1998+A1:2000+A2:2003 Class A
EN 55024:1998+A1:2001+A2:2003
EN 61000-3-2:2000+A2:2005
EN 61000-3-3:1995+A1:2001+A2:2005

This EC-Attestation of Compliance is issued according to the Directive 2004/108/EC relating to electromagnetic compatibility on a voluntary basis. It confirms that the listed apparatus complies with all essential requirements of the EMC directive and applies only to the sample and its technical documentation submitted to TÜV SÜD Product Service GmbH for testing and certification. See also notes overleaf.

Test report no.: EMC-CE-1196

Date, 2007-10-24

216339

CE After preparation of the necessary technical documentation as well as the conformity declaration the required CE marking can be affixed on the product. Other relevant directives have to be observed.

TÜV SÜD Product Service GmbH · Zertifizierstelle · Ridlerstrasse 65 · 80339 München · Germany

EU Declaration of Conformity (EMC)

Type of equipment:	**PCI One Motion Control Board**
Model Name:	**PCI-N804, PCI-N404**
Applicant:	**AJINEXTEK CO.,LTD.**
Address:	**9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea**
Manufacturer:	**AJINEXTEK CO.,LTD.**
Address:	**9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea**

We hereby declare, that all major safety requirements, concerning to CE Mark Directive(93/68/EEC) Electromagnetic Compatibility Directives(2004/108/EC,92/31/EEC) are fulfilled, as laid out in the guideline set down by the member states of the EEC Commission. This declaration is valid for all samples that are part of this declaration, which are manufactured according to the production charts appendix.

The standards relevant for the evaluation of electrical safety requirements are as follows :

EMC : EN 55022:1998+A1+A2:2003, EN 55024:1998+A1+A2:2003
EN 61000-3-2:2000+A2:2005, EN 61000-3-3:1995+A1+A2:2005

1. Certificate of conformity / Test report issued by :

EMC : **EMC Compliance Laboratory**
Certificate No. : **EMC-CE-1196**

2. Technical documentation kept at :

AJINEXTEK CO.,LTD.
which will be made available upon request.

AJINEXTEK CO.,LTD.
9-3 Horim-dong, Dalseo-gu,
Daegu 704-240, Korea

(place and date of issue)

(name and signature of authorized person)

EMC TEST REPORT

Test report no.:	**EMC-CE-1196**
Type of equipment:	**PCI One Motion Control Board**
Model Name:	**PCI-N804, PCI-N404**
Applicant:	**AJINEXTEK CO.,LTD.**
Manufacturer:	**AJINEXTEK CO.,LTD.**
Test standards:	**EN 55022:1998+A1:2000+A2:2003, CLASS A**
	EN 55024:1998+A1:2001+A2:2003
	EN 61000-3-2:2000+A2:2005
	EN 61000-3-3:1995+A1+A2:2005
Testing Laboratory:	**EMC Compliance Ltd.**
Test result:	**Complied**

This product complies with the requirements of the EMC Directive 2004/108/EC.
The results in this report apply only to the sample tested.
This test report shall not be reproduced except in full, without the written Approval of EMC compliance Laboratory.

Date of test: 2007. 10 . 05 ~ 10 . 11

Date of Issue: 2007. 10 . 23

Tested by:  KIM, JIN-SEOK

Approved by:  YOO, SUNG-YUNG

EMC Compliance Ltd.
82-1 Jeil-ri, Yangji-myun, Churingu, Yongin-city, Kyunggi-do 449-825, Korea
TEL: 82 31 336 9919 FAX : 82 31 336 4767
This test report shall not be reproduced except in full, Without the written approval.

Contents

1. Client information 3
2. Laboratory information 4
3. Test system configuration 5
3.1 Operation environment 5
3.2 Measurement Uncertainty 5
3.3 Sample calculation 6
4. Description of EUT 7
4.1 Product description 7
4.2 Peripherals 7
4.3 Operating conditions 8
4.4 Used cables 8
4.5 E.U.T. test configuration 9
5. Summary of test results 10
5.1 Modification to the E.U.T. 10
5.2 Standards & results 10
5.3 Performance criterion 11
6. Test results 12
6.1 Conducted Emission 12
6.2 Radiated emission 19
6.3 Electrostatic Discharge 23
6.4 Radio Frequency Electromagnetic Fields 26
6.5 Electric Fast Transient/BURST 28
6.6 Surge 30
6.7 Conducted Immunity 32
6.8 Dips and Interruptions 34
6.9 Harmonics 35
6.10 Flicker 42
7. Test photographs 45
8. E.U.T. photographs 56

This test report shall not be reproduced except in full, Without the written approval.

1. Client information

Applicant:	**AJINEXTEK CO.,LTD.**
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone number:	+82-53-593-3700
Facsimile number:	+82-53-593-3703

Manufacturer:	**AJINEXTEK CO.,LTD.**
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone number:	+82-53-593-3700
Facsimile number:	+82-53-593-3703

This test report shall not be reproduced except in full, Without the written approval.

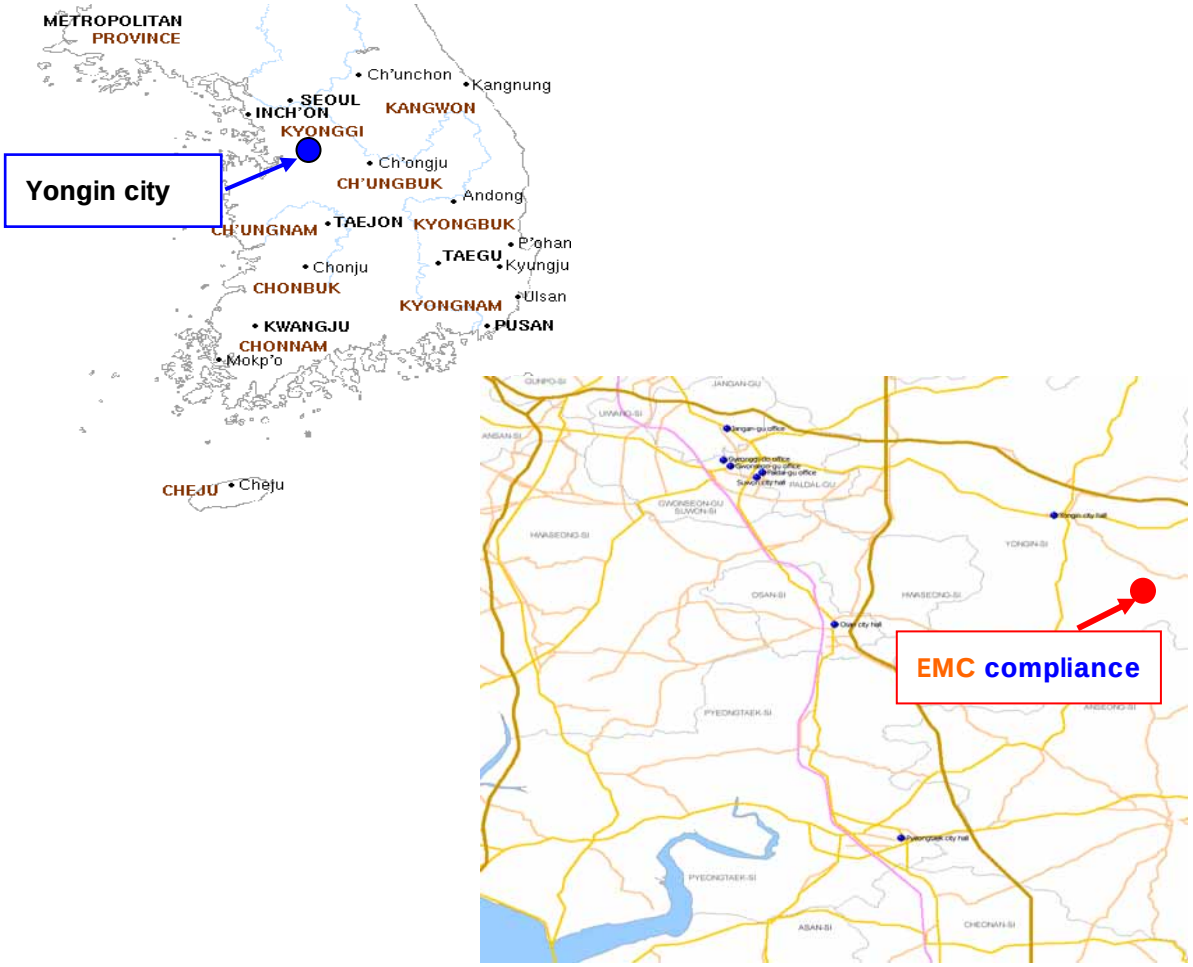
2. Laboratory information

Address

EMC compliance Ltd.
82-1 Jeil-ri, Yangji-myun, Churingu, Yongin-city, Kyunggi-do 449-825, Korea
Telephone Number : 82 31 336 9919
Facsimile Number : 82 31 336 4767

CBTL Testing Laboratory
FCC Filing No.: 793334
FCC CAB.: KR0040
VCCI Registration No. : C-1713, R-1606, T-258
KOLAS NO.: 231

SITE MAP



This test report shall not be reproduced except in full, Without the written approval.

3. Test system configuration

3.1 Operation environment

	Temperature	Humidity	Pressure
OATS :	21 °C	51 %	-
Shielded room :	21 °C	51 %	-
Immunity area :	21 °C	49 %	1000 hPa

Test site

These testing items were performed following locations;

Shielded Room : Conducted Emission
OATS (10m) : Radiated Emission
Immunity area : ESD,RS, EFT/Burst, Surge, CS, Dip/Interruption, Harmonics, Flicker

3.2 Measurement Uncertainty

All measurements involve certain levels of uncertainties, especially in field of EMC. The factors contributing to uncertainties are test receiver, cable loss, antenna factor calibration, Antenna directivity, antenna factor variation with height, antenna phase center variation, antenna frequency interpolation, measurement distance variation, site imperfection, mismatch, and system repeatability. Based on NIS 80, 81, the measurement uncertainty level with a 95% confidence level was applied.

This test report shall not be reproduced except in full, Without the written approval.

3.3 Sample calculation

Conducted Emission

The field strength is calculated by adding the LISN factor, cable loss from the measured reading.
The sample calculation is as follows :
FS = MR + LF + CL
MR = Meter Reading
LF = LISN Factor
CL = Cable Loss
If MR is 30dB, LISN Factor 1dB, CL 1dB
The result (MR) is
30 + 1 + 1 = 32dBuV

Radiated emission

The field strength is calculated adding the antenna Factor, cable loss and, Antenna pad adding, subtracting the amplifier gain from the measured reading.
The sample calculation is as follows :
FS = MR + AF + CL + AT -AG
MR = Meter Reading
AF = Antenna Factor
CL = Cable Loss
AP = Antenna Pad
AG=Amplifier Gain
If MR is 30dB, AF 12dB, CL 5dB, AP 10dB, AG 35dB
The result (MR) is
30 + 12 + 5 + 10 –35 = 22dBuV/m

This test report shall not be reproduced except in full, Without the written approval.

4. Description of EUT

4.1 Product description

Applicant:	AJINEXTEK CO.,LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Manufacturer:	AJINEXTEK CO.,LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Type of equipment:	PCI One Motion Control Board
Basic Model:	PCI-N804, PCI-N404
Serial number:	-
Testing voltage	230Vac, 50Hz
Rating:	230Vac, 50Hz

4.2 Peripherals

Description	Model / Part #	Serial number	Manufacturer
PC	-	-	ADVANTECH
PS/2 keyboard	SPR-8695I	6CEZ052658	SEJIN ELECTORN
PS/ 2 mouse	-	-	
T68-PR	-	-	-

This test report shall not be reproduced except in full, Without the written approval.

4.3 Operating conditions

- Enterprise offer program Monitoring mode

4.4 Used cables

- PCI-N804

Start		END		Cable Spec.	
Name	**I/O Port**	**Name**	**I/O Port**	**Length**	**Shield**
EUT	SCSI cable 1	T-68-PR	SCSI cable 1	2.0	Shield
	SCSI cable 2		SCSI cable 2	2.0	Shield
	SCSI cable 3		SCSI cable 3	2.0	Shield
	SCSI cable 4		SCSI cable 4	2.0	Shield

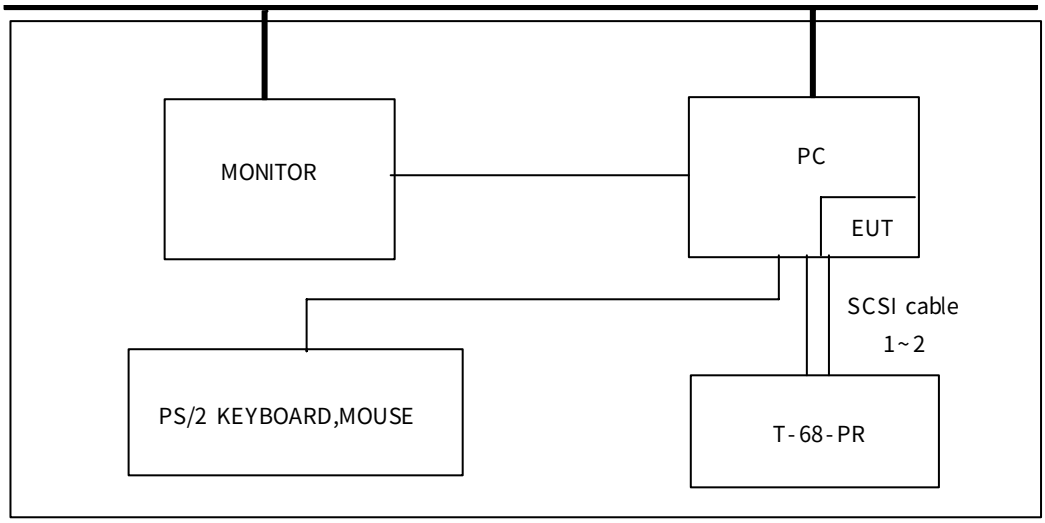
- PCI-N404

Start		END		Cable Spec.	
Name	**I/O Port**	**Name**	**I/O Port**	**Length**	**Shield**
EUT	SCSI cable 1	T-68-PR	SCSI cable 1	2.0	Shield
	SCSI cable 2		SCSI cable 2	2.0	Shield

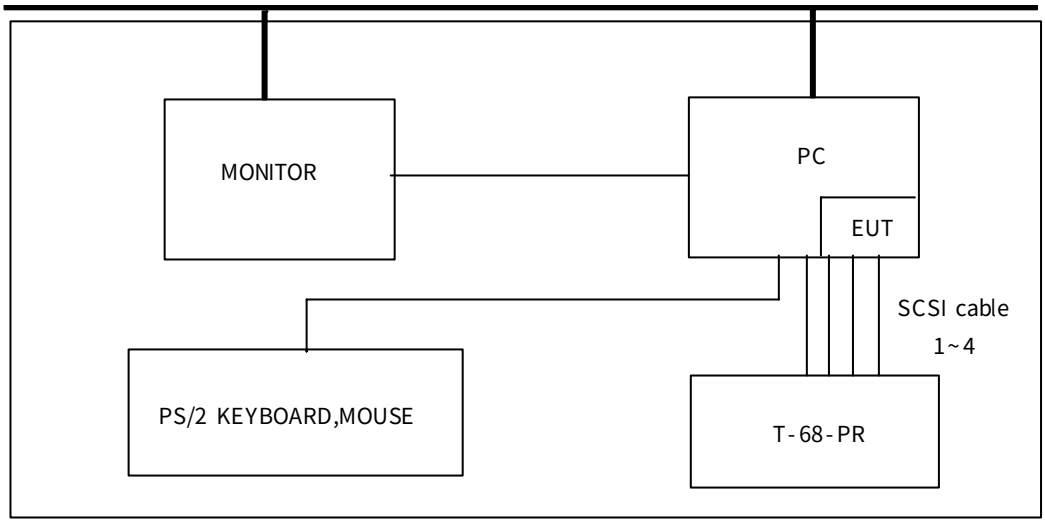
This test report shall not be reproduced except in full, Without the written approval.

4.5 E.U.T. test configuration

- PCI-N804



- PCI-N404



This test report shall not be reproduced except in full, Without the written approval.

5. Summary of test results

5.1 Modification to the E.U.T.

None

5.2 Standards & results

The following standards have been applied:

EN 55022:1998+A1:2000+A2:2003

Information technology equipment – Radio disturbance characteristics - Limits and methods of measurement

Test items	Result
Conducted emission	Pass
Radiated emission	Pass

EN 55024:1998+A1:2001+A2:2003

Information technology Equipment – Immunity characteristics - Limits and Methods of measurement

Test items	Test methods	Result
Electrostatic discharge	EN 61000-4-2:1995+A1:1998+A2:2001	Pass
Electromagnetic field	EN 61000-4-3:1995+A1:1998+A2:2001	Pass
Electric fast transients	EN 61000-4-4:1995+A1:2001	Pass
Surge	EN 61000-4-5:1995+A1:2001	Pass
Conducted Immunity	EN 61000-4-6:1995+A1:2001	Pass
Voltage dip/interruption	EN 61000-4-11:1994+A1:2001	Pass

EN 61000-3-2:2000+A2:2005

Limits for harmonic-current emissions (equipment input current up to including 16A per phase)

Test items	Test methods	Result
Harmonics	EN 61000-3-2:2000+A2:2005	Pass

EN 61000-3-3:1995+A1:2001+A2:2005

Limitation of voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16A per phase and not subject to conditional connection

Test items	Test methods	Result
Flicker	EN 61000-3-3: 1995+A1:2001+A2:2005	Pass

This test report shall not be reproduced except in full, Without the written approval.

5.3 Performance criterion

Performance criterion A: The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion B: After the test, The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test.
If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as untended.

Performance criterion C: Loss of function is allowed, provided the function is self-recoverable or can be restored by the operating of the controls by the user In accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

This test report shall not be reproduced except in full, Without the written approval.

6. Test results

6.1 Conducted Emission

6.1.1 Measurement procedure

Mains

The measurements were performed in a shielded room. EUT was placed on a non-metallic table height of 0.8 m above the reference ground plane. The rear of table was located 0.4 m to the vertical conducted plane. EUT was power through the LISN, which was bonded to the ground plane. The LISN power was filtered. Each EUT power lead, except ground (safety) lead, was individually connected through a LISN to input power source. All I.O cables are positioned to simulate typical actual usage according to the test standard. Both lines of power cord, hot and neutral, were measured.

6.1.2 Used equipments

Equipment	Model	Serial No.	Makers	Next Cal. Date	Used
Test Receiver	ESHS10	843276/003	R&S	08.05.22	☒
LISN	ESH3-Z5	100267	R&S	08.07.02	☒
LISN	L2-16A	0000J10705	PMM	-	☒

6.1.3 Measurement uncertainty

Conducted emission measurement : (k=2, 95%)
9kHz-150 kHz : ± 3.263 [dB]
150kHz-30 MHz : ± 2.960 [dB]

This test report shall not be reproduced except in full, Without the written approval.

6.1.4 Test data

-PCI-N804

Frequency	Correction Factor		Line	Quasi-peak			Average		
				Limit	Reading	Result	Limit	Reading	Result
[MHz]	LISN	Cable		[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]
0.201	0.10	0.2	N	79.00	34.78	35.08	66.00	28.65	28.95
0.204	0.23	0.2	H	79.00	35.42	35.85	66.00	35.85	36.28
0.270	0.23	0.2	H	79.00	31.19	31.62	66.00	23.37	23.80
0.339	0.10	0.2	N	79.00	30.51	30.81	66.00	26.82	27.12
0.405	0.10	0.3	N	79.00	35.47	35.87	66.00	32.93	33.33
0.471	0.22	0.3	H	79.00	33.28	33.80	66.00	30.48	31.00
0.675	0.25	0.2	H	73.00	36.72	37.17	60.00	34.89	35.34
0.744	0.25	0.2	H	73.00	34.56	35.01	60.00	32.52	32.97
0.945	0.25	0.3	H	73.00	36.72	37.27	60.00	34.54	35.09
0.945	0.13	0.3	N	73.00	36.84	37.27	60.00	34.54	34.97
1.011	0.25	0.3	H	73.00	35.74	36.29	60.00	33.49	34.04
10.160	0.35	0.5	N	73.00	25.56	26.41	60.00	21.41	22.26

• Note. QP = Quasi-Peak, AV= Average

• Loss = LISN Loss + Cable Loss

• Measurement time : 1 s

This test report shall not be reproduced except in full, Without the written approval.

PCI-N804

Frequency	Correction Factor		Line	Quasi-peak			Average		
				Limit	Reading	Result	Limit	Reading	Result
[MHz]	LISN	Cable		[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]	[dBuV]
0.204	0.23	0.2	H	79.00	35.74	36.17	66.00	31.04	31.47
0.270	0.23	0.2	H	79.00	31.33	31.76	66.00	26.01	26.44
0.339	0.22	0.2	H	79.00	29.17	29.59	66.00	24.27	24.69
0.405	0.10	0.3	N	79.00	35.71	36.11	66.00	33.13	33.53
0.471	0.22	0.3	H	79.00	33.24	33.76	66.00	30.41	30.93
0.675	0.12	0.2	N	73.00	36.74	37.06	60.00	34.98	35.30
0.741	0.25	0.2	H	73.00	35.02	35.47	60.00	33.06	33.51
0.945	0.13	0.3	N	73.00	36.70	37.13	60.00	34.45	34.88
1.011	0.13	0.3	N	73.00	35.91	36.34	60.00	33.63	34.06
1.215	0.13	0.3	N	73.00	34.56	34.99	60.00	31.65	32.08

• Note. QP = Quasi-Peak, AV= Average

• Loss = LISN Loss + Cable Loss

• Measurement time : 1 s

This test report shall not be reproduced except in full, Without the written approval.

-PCI-N804

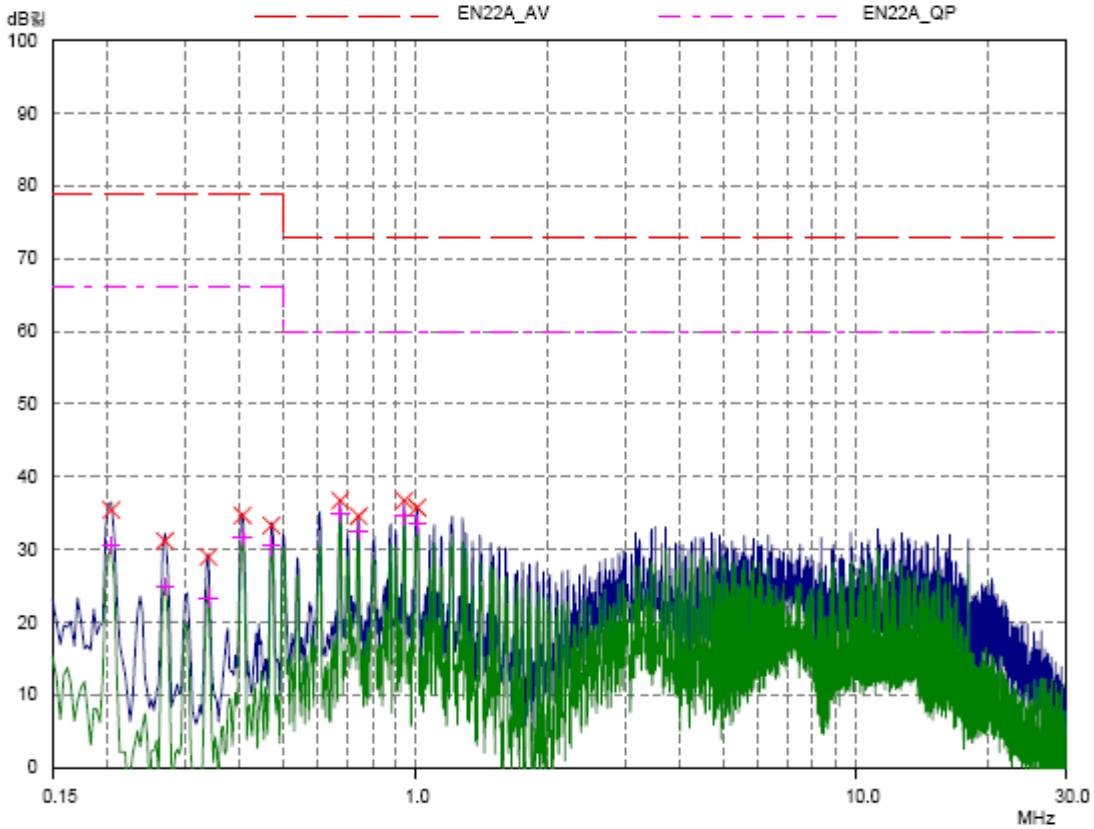
EUT: PCI-N804
Manuf: AZINEXTEK
Op Cond: H
Operator:
Test Spec: EN55022 Class A Conducted Emission
Comment:

Result File: 804h.dat : 804_H

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	2msec	Auto	OFF	60dB

Final Measurement: Detectors: X QP / + AV
Meas Time: 1sec
Peaks: 8
Acc Margin: 25 dB



This test report shall not be reproduced except in full, Without the written approval.

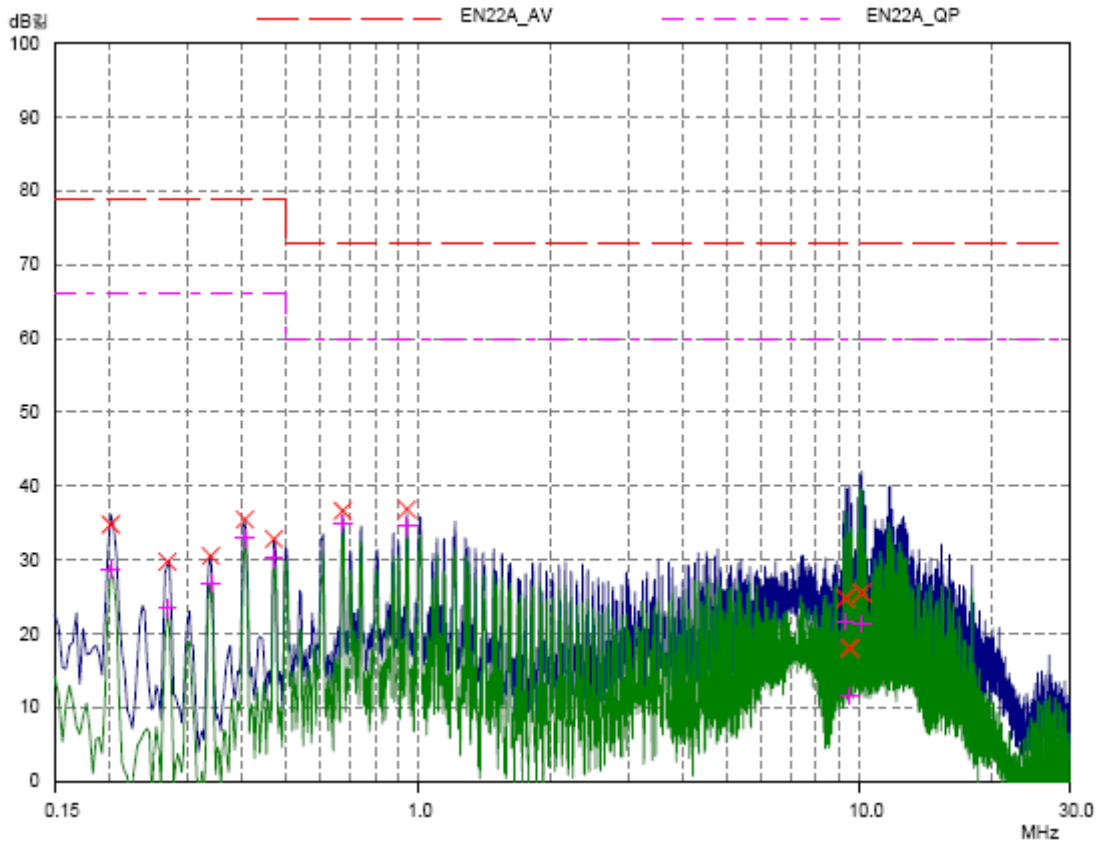
EUT: PCI-N804
Manuf: AZINEXTEK
Op Cond: N
Operator:
Test Spec: EN55022 Class A Conducted Emission
Comment:

Result File: 804n.dat : 804_N

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	2msec	Auto	OFF	60dB

Final Measurement: Detectors: X QP / + AV
Meas Time: 1sec
Peaks: 8
Acc Margin: 25 dB



This test report shall not be reproduced except in full, Without the written approval.

-PCI-N404

EUT:	PCI-N404
Manuf:	AZINEXTEK
Op Cond:	H
Operator:	
Test Spec:	EN55022 Class A Conducted Emission
Comment:	

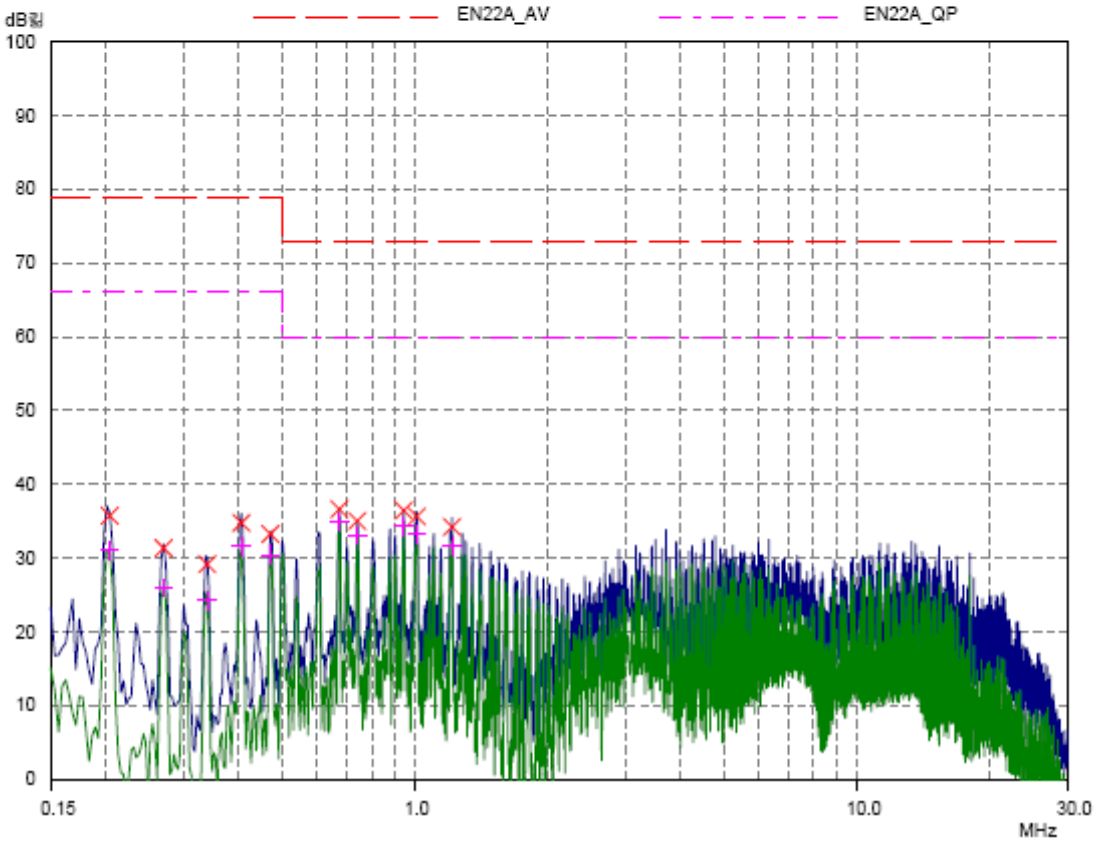
Result File: 404h.dat : 404_H

Scan Settings (2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	2msec	Auto	OFF	60dB

Final Measurement:

Detectors:	X QP / + AV
Meas Time:	1sec
Peaks:	8
Acc Margin:	25 dB

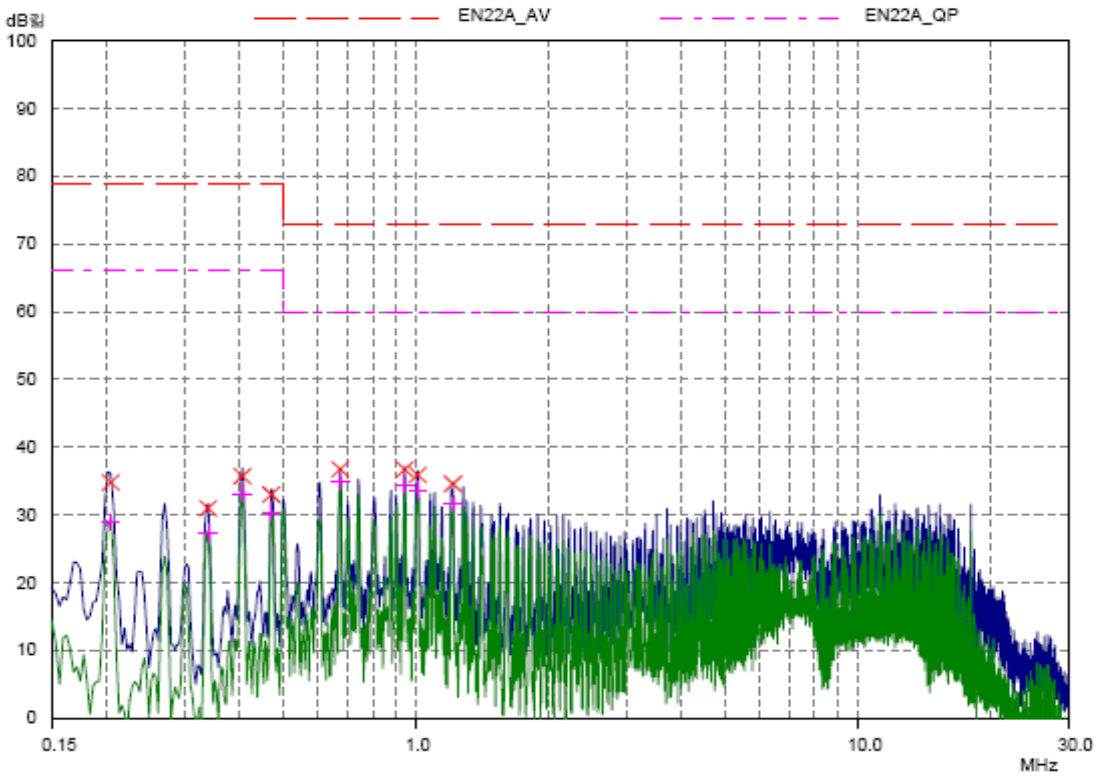


This test report shall not be reproduced except in full, Without the written approval.

EUT:	PCI-N404
Manuf:	AZINEXTEK
Op Cond:	N
Operator:	
Test Spec:	EN55022 Class A Conducted Emission
Comment:	
Result File:	404n.dat : 404_N
Scan Settings	(2 Ranges)

Frequencies			Receiver Settings					
Start	Stop	Step	IF BW	Detector	M-Time	Atten	Preamp	OpRge
150kHz	3MHz	3kHz	10kHz	PK+AV	5msec	Auto	OFF	60dB
3MHz	30MHz	10kHz	10kHz	PK+AV	2msec	Auto	OFF	60dB

Final Measurement:	Detectors:	X QP / + AV
	Meas Time:	1sec
	Peaks:	8
	Acc Margin:	25 dB



6.1.5 Result

Minimum limit margin is 24.91 dB at 0.945 MHz. (CPI-N804)

Minimum limit margin is 24.70 dB at 0.675 MHz. (CPI-N404)

This test report shall not be reproduced except in full, Without the written approval.

6.2 Radiated emission

6.2.1 Measurement procedure

Mains

A pretest was performed at 3m distance in a semi-anechoic chamber for searching correct frequency. The final test was done at a 10m open area test site with a quasi-peak detector. EUT was placed on a non-metallic table height of 0.8m above the reference ground plane. Cables were folded back and forth forming a bundle 0.3m to 0.4m long and were hanged at a 0.4m height to the ground plane.
Cables connected to EUT were fixed to cause maximum emission. Test was made with the antenna positioned in both the horizontal and vertical planes of polarization.
The measurement antenna was varied in height above the conducting ground plane to obtain the maximum signal strength.

6.2.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next cal. date	Used
Test Receiver	ESCI	100001	R&S	07.10.31	☒
TRILOG broadband antenna	VULB 9160	3228	Schwarzbeck Mess-Electronik	08.02.12	☒
Antenna Mast	A109	N/A	DEAIL	-	☒
Turn Table	TS14	N/A	DEAIL	-	☒
10m OATS	-	-	EMC Compliance	-	☒

6.2.3 Measurement uncertainty

Radiated Emission measurement : (k=2, 95%)
30-300 MHz ; 3 m: ±3.721 [dB], 10 m: +3.706, -3.707 [dB]
300-1000 MHz ; 3 m: ±3.818 [dB], 10 m: ±3.802 [dB]

This test report shall not be reproduced except in full, Without the written approval.

6.2.4 Test data

- PCI-N804

Frequency	Reading	Pol.	Height	angle	Correction Factor		Limits	Result	Margin
[MHz]	[dBuV/m]		[m]		Antenna	Cable	[dBuV/m]	[dBuV/m]	[dB]
31.70	13.9	V	1.0	312	11.21	1.25	40.0	26.36	13.64
54.195	7.9	V	1.1	185	11.55	1.63	40.0	21.09	18.91
144.02	13.7	H	3.5	268	12.41	2.41	40.0	28.51	11.49
231.36	12.9	H	4.0	168	10.75	3.00	47.0	26.65	20.35
500.23	5.2	H	3.0	360	17.46	4.52	47.0	27.18	19.82
564.012	7.6	V	3.0	17	18.78	4.78	47.0	31.16	15.84
599.99	5.0	V	4.0	360	19.68	4.90	47.0	29.58	17.42
600.015	5.4	H	2.2	352	19.71	4.90	47.0	30.01	16.99

* Receiving Antenna Mode : Horizontal, Vertical

* 10 m OATS

* Note : Reading = Test Receiver meter,

P= Polarization → H = Horizontal, V = Vertical

Result = Field Strength (Antenna factor + Cable factor + Reading)

- PCI -N404

Frequency	Reading	Pol.	Height	angle	Correction Factor		Limits	Result	Margin
[MHz]	[dBuV/m]		[m]		Antenna	Cable	[dBuV/m]	[dBuV/m]	[dB]
86.251	16.9	H	4.0	360	8.02	1.97	40.0	26.89	13.11
91.15	18.0	V	1.4	169	8.33	2.06	40.0	28.38	11.62
91.52	19.3	H	3.5	357	8.33	2.06	40.0	29.68	10.32
100.70	16.4	H	3.0	184	9.18	2.13	40.0	27.71	12.29
132.50	9.7	H	3.6	148	11.76	2.34	40.0	23.80	16.20
299.898	3.0	V	2.5	12	13.00	3.41	47.0	19.41	27.59
300.125	13.4	H	2.7	296	13.03	3.41	47.0	29.84	17.16

* Receiving Antenna Mode : Horizontal, Vertical

* 10 m OATS

* Note : Reading = Test Receiver meter,

P= Polarization → H = Horizontal, V = Vertical

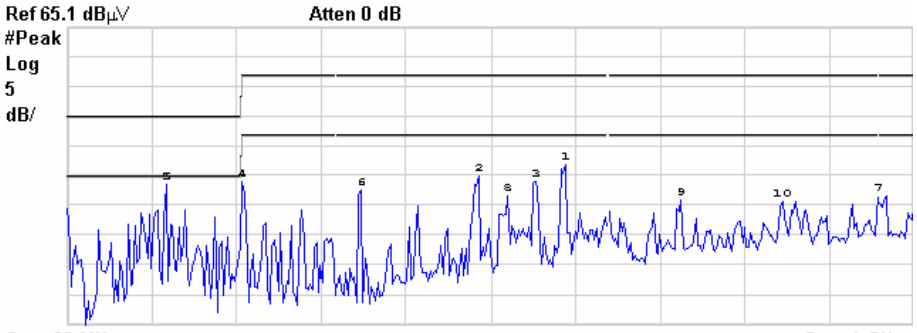
Result = Field Strength (Antenna factor + Cable factor + Reading)

This test report shall not be reproduced except in full, Without the written approval.

***3m Chamber Prescan Data**

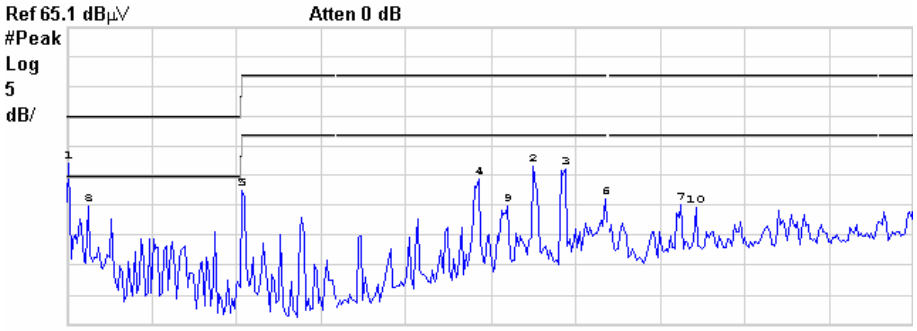
- REC MODE(PCI-N804)

-Horizontal



Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	599.9 MHz	41.9 dBµV	6	367.1 MHz	37.67 dBµV
2	500.5 MHz	40.02 dBµV	7	958.8 MHz	36.71 dBµV
3	565.9 MHz	39.05 dBµV	8	534.4 MHz	36.53 dBµV
4	231.3 MHz	38.95 dBµV	9	733.3 MHz	35.85 dBµV
5	144.0 MHz	38.49 dBµV	10	849.7 MHz	35.78 dBµV

-Vertical



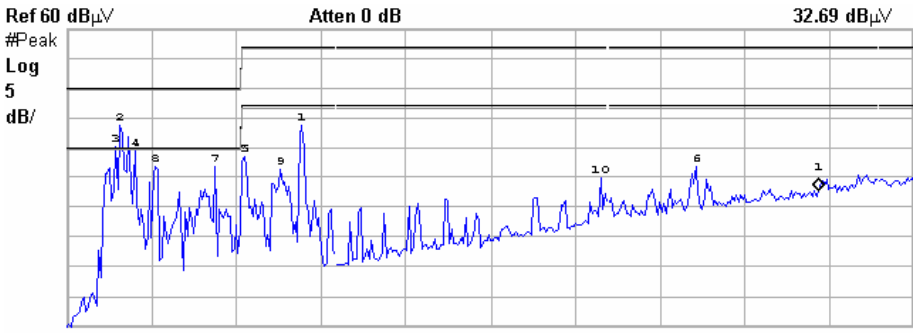
Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	32.4 MHz	42.13 dBµV	6	646.0 MHz	36.1 dBµV
2	563.5 MHz	41.73 dBµV	7	733.3 MHz	35.18 dBµV
3	599.9 MHz	41.29 dBµV	8	54.3 MHz	34.98 dBµV
4	500.5 MHz	39.62 dBµV	9	534.4 MHz	34.92 dBµV
5	231.3 MHz	37.6 dBµV	10	750.2 MHz	34.67 dBµV

This test report shall not be reproduced except in full, Without the written approval.

***3m Chamber Prescan Data**

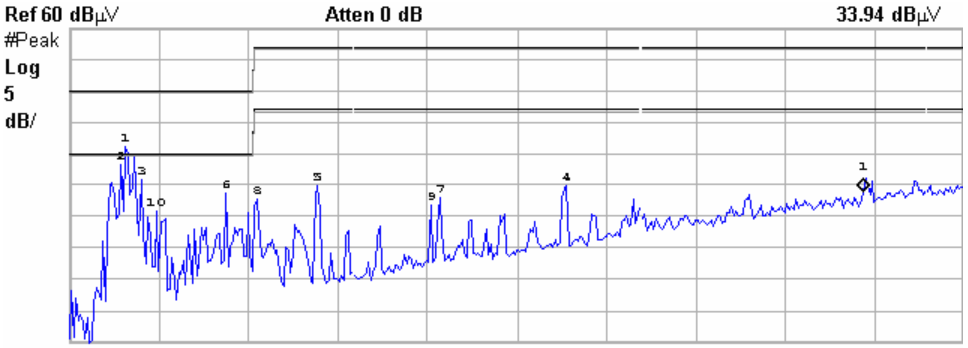
- REC MODE(PCI -N404)

-Horizontal



Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	299 MHz	43.81 dBµV	6	750 MHz	36.83 dBµV
2	91 MHz	43.64 dBµV	7	200 MHz	36.82 dBµV
3	86 MHz	40.16 dBµV	8	132 MHz	36.71 dBµV
4	108 MHz	39.47 dBµV	9	275 MHz	36.2 dBµV
5	234 MHz	38.4 dBµV	10	641 MHz	34.77 dBµV

-Vertical



Pk	X Axis	Amplitude	Pk	X Axis	Amplitude
1	91 MHz	41.06 dBµV	6	200 MHz	33.61 dBµV
2	86 MHz	38.14 dBµV	7	433 MHz	32.92 dBµV
3	108 MHz	35.93 dBµV	8	234 MHz	32.66 dBµV
4	568 MHz	34.77 dBµV	9	423 MHz	31.83 dBµV
5	299 MHz	34.77 dBµV	10	125 MHz	30.87 dBµV

6.2.5. Result

Minimum limit margin is 11.49 dB at 144.02 MHz. (CPI-N804)

Minimum limit margin is 10.32 dB at 91.52 MHz. (CPI-N404)

This test report shall not be reproduced except in full, Without the written approval.

6.3 Electrostatic Discharge

6.3.1 Measurement procedure

A ground reference plane was located on the floor, and connected to earth via a low impedance connection. The return cable of the ESD generator was connected to the reference plane. In case of floor standing equipment, EUT was placed on the reference plane on 0.1 m of insulating Support. In case of table top equipment, EUT was placed on a wooden table 0.8m above the reference grounded floor. A horizontal coupling plane(HCP) was placed on the table, and Connected to the reference plane via a 470㏀ resistor located in each end (0.5mm insulating support between EUT and HCP). In both cases a vertical coupling plane(VCP) OF 0.5 X 0.5m was located 10cm from the EUT's sides. The VCP was connected to the reference plane in the same matter as the HCP.

6.3.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. Date	Used
ESD Tester	ESS-100L	5099C02066	NOISE KEN	08.06.19	☒
HCP	-	-	-	-	☒
VCP	-	-	-	-	☒

This test report shall not be reproduced except in full, Without the written approval.

6.3.3 Test Data

Test Specification : EN61000-4-2

Kind of Discharges

☒ Contact Discharge (Direct Discharge)

☐ Air Discharge

☒ HCP / VCP (Indirect Discharge)

Discharge Voltages

☒ Contact Discharge : ± 2, 4kV

☐ Air Discharge :

Discharge Impedance

☒ 330 Ω/150pF ☐ 2KΩ/330pF

Number Of Discharge

☒ Number of discharges per point, for each voltage and polarity
: 50 (Interval between discharges : ≥ 1s)

Test point (Please refer to attached photograph.)

▪Contact Discharge : Bleraket, Port

▪Air Discharge :

Test Results

☒ Complied ☐ Not complied

Comment :

-There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

Electrostatic Discharge (Test Point)

Air discharge →
Contact discharge →

- PCI -N804



- PCI-N404



This test report shall not be reproduced except in full, Without the written approval.

6.4 Radio Frequency Electromagnetic Fields

6.4.1 Measurement procedure

The test was performed at 3m full anechoic chamber.
For floor standing equipment, the EUT was standing on the floor.
For tabletop equipment, the EUT was located on a wooden table 0.8m above the floor.
The EUT was tested all sides, horizontal and vertical polarization.
The field uniformity was calibrated for 1V/m, 3V/m, 10V/m.

6.4.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Power meter	PM2002	302852	AR	08.05.03	☒
Field monitor	SI-300	-	TDK	-	☒
Power sensor (with adapter)	PH2000	303224	AR	08.05.03	☒
Power sensor (with adapter)	PH2000	303222	AR	08.05.03	☒
Directional coupler	DC6180	303976	AR	08.05.03	☒
Amplifier	150W1000M2	303843	AR	08.05.03	☒
Amplifier	60S1G3M2	320444	AR	08.04.21	☒
Signal generator	E4421B	GB40052295	AGILENT	07.10.24	☒
BiconiLog Ant.	LPDA-0803	130269	ETS	08.03.17	☒
Directional Coupler	DC7144M1	320279	AR	08.01.18	☒

6.4.3 Measurement uncertainty

Radio Frequency Electromagnetic Fields : ± 1.0556 [dB](k=2, 95%)

This test report shall not be reproduced except in full, Without the written approval.

6.4.4 Test Data

Test Specification : EN 61000-4-3

Frequency Range

☒ 80MHz – 1000MHz ☐ 900 MHz ± 5MHz ☐ 26MHz - 500MHz

Test level

☐1V/m ☒ 3V/m ☐ 10V/m

Modulation

☒ AM : 1kHz, 80%

☐ PM :

Frequency step

☒ log 1% step ☐ log 3% step ☐ log 5% step

Dwell Time

☒ 3 s ☐ 2 s ☐ 1 s

Test point

☒ Front

☒ Rear

☒ Left

☒ Right

Test Results

☒ Complied ☐ Not complied

Comment :

- There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.5 Electric Fast Transient/BURST

6.5.1 Measurement procedure

A ground reference plane was located on the floor.
EFT generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.8 m wooden table.
For tabletop equipment, EUT was placed on a wooden table(0.8m) above the reference plane.

6.5.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
EFT/B Tester	UCS 500 M6	V0545100858	EM TEST	08.01.03	☒
Capacitive coupling clamp	N/A	N/A	EM TEST	-	☐

This test report shall not be reproduced except in full, Without the written approval.

6.5.3 Test Data

Test Specification : EN 61000-4-4

Coupling

☒ Power ☐ Signal Lines ☐ Telecommunication line

Test level

☒ Power : ±1 kV

☐ Signal Line :

☐ Tel. line :

Test mode

- Power : L1, L2, PE, L1+L2, L1+PE, L2+PE, L1+L2+PE

-Signal Line :

Burst frequency : 5 kHz, 5/50 ns

Coupling Time : > 60 s

Test Results

☒ Complied ☐ Not complied

Comment :

- There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.6 Surge

6.6.1 Measurement procedure

A ground reference plane was located on the floor.
SURGE generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.8 m wooden table.
For tabletop equipment, EUT was placed on a wooden table(0.8m) above the reference plane.

6.6.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Surge Generator	UCS 500 M6	V0545100858	EM TEST	08.01.03	☒
CDN	CNV 508 S2	0402-01	EM TEST	-	☐

This test report shall not be reproduced except in full, Without the written approval.

6.6.3 Test Data

Test Specification : EN 61000-4-5

Coupling

☒ Power ☐ Signal Line ☐ Telecommunication line

Test level

☒ Power : L-N : ± 1 kV
L-PE, N-PE : ± 2 kV

☐ Signal Line :

☐ Tel. line :

Test mode

- AC Power : L-N, L-PE, N-PE

-Signal Line :

Coupling Impedance

☐ 40Ω+0.5μF ☐ 40Ω ☒ 10Ω+9μF ☒ 18μF

Coupling Time : ＞ 1 min

Number of Surge : 5

Angle : 0, 90, 180, 270

Test Results

☒ Complied ☐ Not complied

Comment :

-There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.7 Conducted Immunity

6.7.1 Measurement procedure

A ground reference plane was located on the floor.
For tabletop equipment, the test was performed on a ground reference plane on a 0.8m wooden table.
The EUT was isolated 0.1 m isolating support.
The ground plane was connected to floor reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.8 m wooden table.
This test were Performed using CDN for mains, clamp for signal and injection probe.

6.7.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
CS generator	CWS 500 C S1	V0635101750	EM TEST	2007.11.02	☒
CDN	CDN M2/M3	0906-12	EM TEST	2007.11.02	☒
Attenuator	INA2070-1	2054	Schaffner	-	☒
EM Clamp	KEMZ 801	17643	Schaffner	-	☐
EM Clamp	KEMA 801	17899	Scahffner	-	☐
Current probe	MD720	W1345167/M6/0068	Schaffner	-	☐

This test report shall not be reproduced except in full, Without the written approval.

6.7.3 Test Data

Test Specification : EN 61000-4-6

Frequency Range

☒ 150 kHz – 80MHz ☐ 150 kHz - 230MHz ☐150 kHz - 500MHz

Test point: Power

Coupling

☒ Power : CDN

☐ Signal :

☐ Tel. line :

Test level

☐ 1V ☒ 3V ☐ 10V

Modulation

☒ AM : 1kHz, 80%

☐ PM : 1Hz (0.5 s ON : 0.5 s OFF)

Frequency step

☒ log 1% step ☐ log 3% step ☐ log 5% step

Dwell Time

☒ 3 s ☐ 2 s ☐ 1 s

Test Results

☒ Complied ☐ Not complied

Comment :

-There was no change of operation status during above testing.

This test report shall not be reproduced except in full, Without the written approval.

6.8 Dips and Interruptions

6.8.1 Measurement procedure

The dips/interruption test is only applicable to AC mains.
The dips/interruptions were applied at zero crossing.

6.8.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
dips/interruption Tester	UCS 500 M6	V0545100858	EM TEST	08.01.03	☒

6.8.3 Test data

Test specification : EN 61000-4-11

Test data

- 230Vac

Test Level (%UT)	Dip/Int. (%UT)	Duration /Period	Phase (°)	Count number	Result
0%	100%	0.5 Period	0 / 180	3T	A
70%	30%	25 Period	0 / 180	3T	A
0 %	100 %	250 Period	0 / 180	3T	C

Test results

☒ Complied ☐ Not complied

Comment :

- The power of EUT is off during the interruption test. After the test, EUT is getting back to normal operation. (250 Period)

This test report shall not be reproduced except in full, Without the written approval.

6.9 Harmonics

6.9.1 Measurement procedure

The equipment is supplied in series with shunt(s) Rm or current transformer(s) from a source having the same nominal voltage and frequency as the rated supply voltage and frequency of the equipment.

Measurements shall be made under normal load, or conditions for adequate heat discharge, and under normal operating conditions.

User's operation controls or automatic programmers shall be set to produce the maximum harmonic component, for each successive harmonic component in turn. For the purpose of harmonic current limitation, equipment is classified as follows :

Class A : Equipment not specified in one of the three other Classes shall be considered as Class A equipment.

- Balanced three-phase equipment;
- Household appliances excluding equipment identified as Class D;
- Tools excluding portable tools;
- Dimmers for incandescent lamps;
- Audio equipment.

Class B : Portable tools; Arc welding equipment which is not professional equipment.

Class C : Lighting equipment.

Class D : Equipment having a specified power according to 6.2.2 less than or equal to 600 w,
of the following types:

- Personal computers and personal computer monitors;
- Television receivers.

6.9.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Harmonics/Flicker meter	5001x-CTS-400-413	54984	C.I.	08.05.03	☒
Test site	Immunity area	-	-	-	☒

6.9.3 Test data

- Refer to attached test data

Test results

☒ Complied ☐ Not complied

This test report shall not be reproduced except in full, Without the written approval.

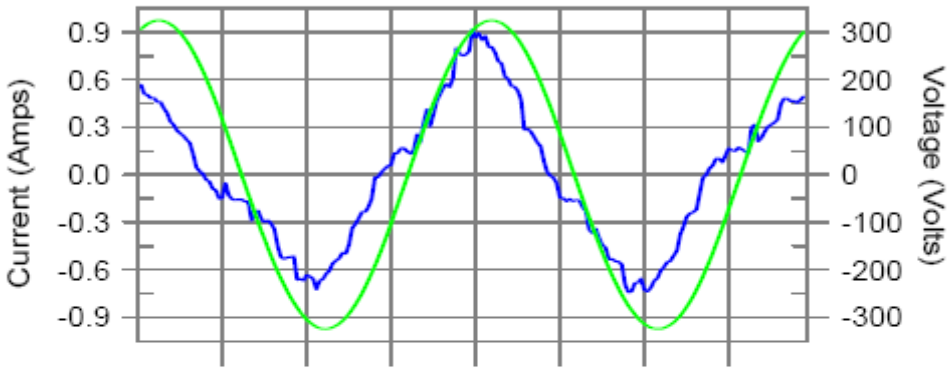
- PCI-N804

Harmonics – Class-A per Ed. 3.0 (2005-11)(Run time) incl. inter-harmonics

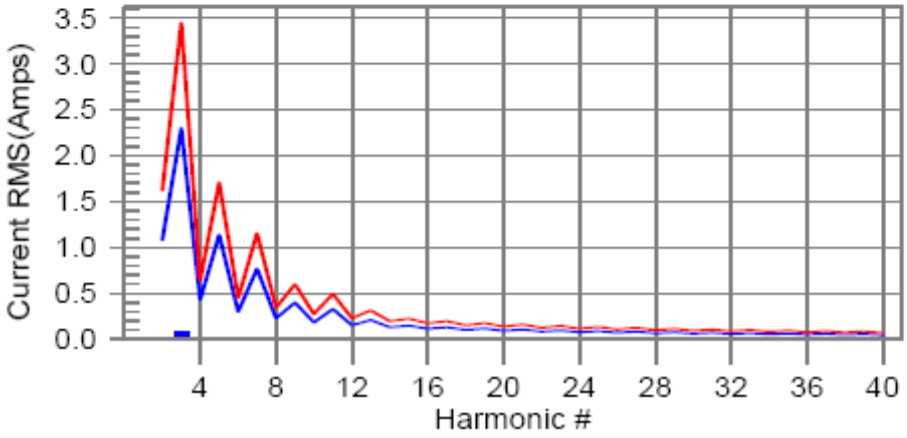
EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-10-01 Start time: 오후 9:24:06 End time: 오후 9:26:47
Test duration (min): 2.5 Data file name: H-000280.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Current & voltage waveforms



Harmonics and Class A limit line European Limits



Test result: Pass Worst harmonic was #15 with 7.49% of the limit.

This test report shall not be reproduced except in full, Without the written approval.

Current Test Result Summary (Run time)

EUT: Equipment Under Test — Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) — Test Margin: 100
Test date: 2007-10-01 — Start time: 오후 9:24:06 — End time: 오후 9:26:47
Test duration (min): 2.5 — Data file name: H-000280.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass — Source qualification: Normal
THC(A): 0.07 — I-THD(%): 21.12 — POHC(A): 0.011 — POHC Limit(A): 0.251
Highest parameter values during test:

V_RMS (Volts):	229.78	Frequency(Hz):	50.00
I_Peak (Amps):	0.935	I_RMS (Amps):	0.470
I_Fund (Amps):	0.439	Crest Factor:	2.192
Power (Watts):	99.8	Power Factor:	0.925

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.005	1.080	0.5	0.022	1.620	1.39	Pass
3	0.069	2.300	3.0	0.075	3.450	2.17	Pass
4	0.001	0.430	0.2	0.003	0.645	0.42	Pass
5	0.011	1.140	1.0	0.012	1.710	0.67	Pass
6	0.001	0.300	0.3	0.002	0.450	0.52	Pass
7	0.004	0.770	0.5	0.006	1.155	0.52	Pass
8	0.001	0.230	0.3	0.003	0.345	0.76	Pass
9	0.007	0.400	1.9	0.008	0.600	1.31	Pass
10	0.001	0.184	0.4	0.004	0.276	1.28	Pass
11	0.004	0.330	1.2	0.005	0.495	0.99	Pass
12	0.001	0.153	0.9	0.004	0.230	1.95	Pass
13	0.016	0.210	7.7	0.017	0.315	5.41	Pass
14	0.002	0.131	1.5	0.006	0.197	3.08	Pass
15	0.007	0.150	4.8	0.017	0.225	7.49	Pass
16	0.002	0.115	1.5	0.006	0.173	3.65	Pass
17	0.005	0.132	3.4	0.011	0.199	5.28	Pass
18	0.002	0.102	1.6	0.006	0.153	3.69	Pass
19	0.010	0.118	8.4	0.010	0.178	5.83	Pass
20	0.001	0.092	1.2	0.004	0.138	3.15	Pass
21	0.004	0.107	3.8	0.004	0.161	2.65	Pass
22	0.001	0.084	0.7	0.003	0.125	2.38	Pass
23	0.003	0.098	2.6	0.003	0.147	2.04	Pass
24	0.001	0.077	0.8	0.002	0.115	1.65	Pass
25	0.005	0.090	5.6	0.005	0.135	4.00	Pass
26	0.001	0.071	1.6	0.002	0.106	2.02	Pass
27	0.005	0.083	6.5	0.006	0.125	4.72	Pass
28	0.001	0.066	1.9	0.002	0.099	2.43	Pass
29	0.005	0.078	6.0	0.005	0.116	4.33	Pass
30	0.001	0.061	1.8	0.002	0.092	2.45	Pass
31	0.003	0.073	4.3	0.005	0.109	4.50	Pass
32	0.001	0.058	2.1	0.003	0.086	3.26	Pass
33	0.003	0.068	4.2	0.003	0.102	3.36	Pass
34	0.001	0.054	1.8	0.002	0.081	2.60	Pass
35	0.002	0.064	3.8	0.003	0.096	2.92	Pass
36	0.001	0.051	1.3	0.002	0.077	2.58	Pass
37	0.003	0.061	4.3	0.003	0.091	3.47	Pass
38	0.001	0.048	1.1	0.002	0.073	2.41	Pass
39	0.001	0.058	2.2	0.003	0.087	3.59	Pass
40	0.001	0.046	1.1	0.001	0.069	1.82	Pass

This test report shall not be reproduced except in full, Without the written approval.

Voltage Source Verification Data (Run time)

EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-10-01 Start time: 오후 9:24:06 End time: 오후 9:26:47
Test duration (min): 2.5 Data file name: H-000280.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Highest parameter values during test:

Voltage (Vrms):	229.78	Frequency(Hz):	50.00
I_Peak (Amps):	0.935	I_RMS (Amps):	0.470
I_Fund (Amps):	0.439	Crest Factor:	2.192
Power (Watts):	99.8	Power Factor:	0.925

Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.081	0.460	17.52	OK
3	0.642	2.068	31.05	OK
4	0.079	0.460	17.27	OK
5	0.063	0.919	6.81	OK
6	0.061	0.460	13.27	OK
7	0.058	0.689	8.45	OK
8	0.044	0.460	9.60	OK
9	0.037	0.460	7.95	OK
10	0.032	0.460	6.97	OK
11	0.047	0.230	20.51	OK
12	0.033	0.230	14.44	OK
13	0.040	0.230	17.22	OK
14	0.024	0.230	10.49	OK
15	0.033	0.230	14.57	OK
16	0.034	0.230	14.69	OK
17	0.022	0.230	9.41	OK
18	0.023	0.230	10.04	OK
19	0.028	0.230	12.00	OK
20	0.025	0.230	11.01	OK
21	0.025	0.230	10.91	OK
22	0.014	0.230	5.97	OK
23	0.015	0.230	6.70	OK
24	0.015	0.230	6.49	OK
25	0.014	0.230	6.22	OK
26	0.018	0.230	7.82	OK
27	0.013	0.230	5.47	OK
28	0.013	0.230	5.86	OK
29	0.026	0.230	11.33	OK
30	0.023	0.230	10.23	OK
31	0.015	0.230	6.31	OK
32	0.014	0.230	5.99	OK
33	0.019	0.230	8.11	OK
34	0.020	0.230	8.84	OK
35	0.016	0.230	7.16	OK
36	0.011	0.230	4.71	OK
37	0.013	0.230	5.72	OK
38	0.011	0.230	4.76	OK
39	0.011	0.230	4.87	OK
40	0.011	0.230	4.92	OK

This test report shall not be reproduced except in full, Without the written approval.

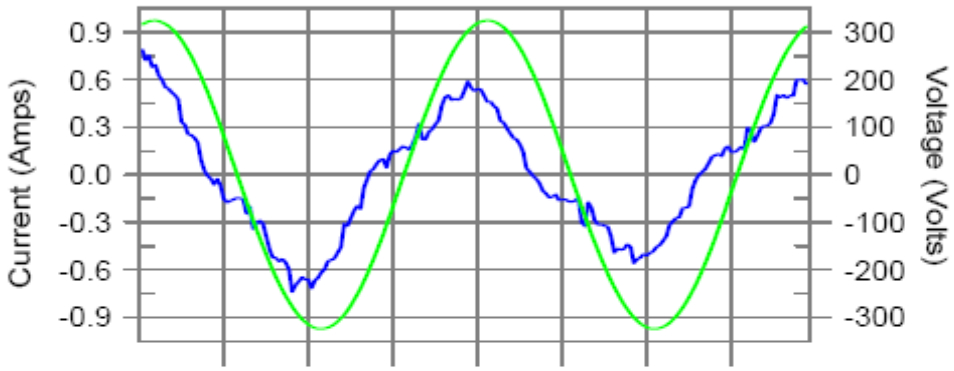
- PCI-N404

Harmonics – Class-A per Ed. 3.0 (2005-11)(Run time) incl. inter-harmonics

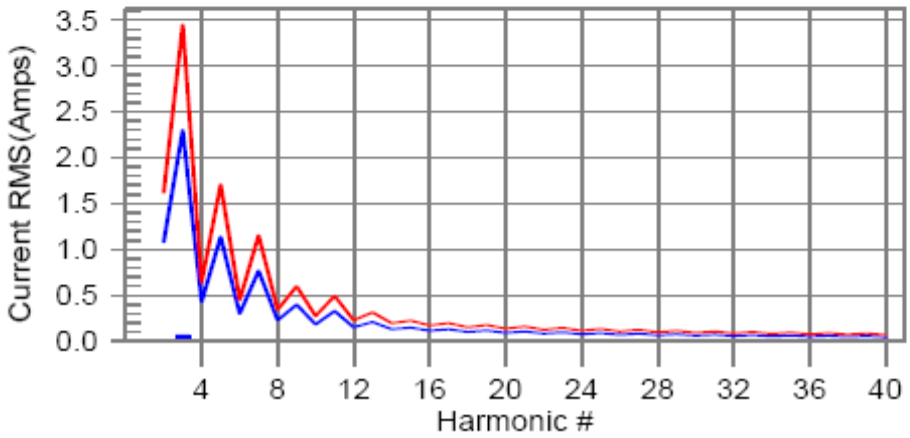
EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-10-01 Start time: 오후 9:20:53 End time: 오후 9:23:34
Test duration (min): 2.5 Data file name: H-000279.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Current & voltage waveforms



Harmonics and Class A limit line European Limits



Test result: Pass Worst harmonic was #19 with 5.78% of the limit.

This test report shall not be reproduced except in full, Without the written approval.

Current Test Result Summary (Run time)

EUT: Equipment Under Test — Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) — Test Margin: 100
Test date: 2007-10-01 — Start time: 오후 9:20:53 — End time: 오후 9:23:34
Test duration (min): 2.5 — Data file name: H-000279.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass — Source qualification: Normal
THC(A): 0.07 — I-THD(%): 21.26 — POHC(A): 0.012 — POHC Limit(A): 0.251
Highest parameter values during test:

V_RMS (Volts):	229.78	Frequency(Hz):	50.00
I_Peak (Amps):	0.799	I_RMS (Amps):	0.375
I_Fund (Amps):	0.356	Crest Factor:	2.183
Power (Watts):	76.2	Power Factor:	0.885

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.004	1.080	0.4	0.006	1.620	0.37	Pass
3	0.069	2.300	3.0	0.070	3.450	2.02	Pass
4	0.001	0.430	0.2	0.001	0.645	0.17	Pass
5	0.011	1.140	1.0	0.011	1.710	0.66	Pass
6	0.001	0.300	0.2	0.001	0.450	0.19	Pass
7	0.004	0.770	0.5	0.004	1.155	0.37	Pass
8	0.001	0.230	0.2	0.001	0.345	0.21	Pass
9	0.008	0.400	1.9	0.008	0.600	1.32	Pass
10	0.001	0.184	0.4	0.001	0.276	0.37	Pass
11	0.004	0.330	1.2	0.004	0.495	0.83	Pass
12	0.001	0.153	0.7	0.001	0.230	0.63	Pass
13	0.017	0.210	7.9	0.017	0.315	5.42	Pass
14	0.002	0.131	1.4	0.002	0.197	1.22	Pass
15	0.007	0.150	4.8	0.008	0.225	3.63	Pass
16	0.002	0.115	1.4	0.002	0.173	1.25	Pass
17	0.004	0.132	3.3	0.005	0.199	2.40	Pass
18	0.001	0.102	1.1	0.002	0.153	1.22	Pass
19	0.010	0.118	8.4	0.010	0.178	5.78	Pass
20	0.001	0.092	1.2	0.001	0.138	1.08	Pass
21	0.004	0.107	3.9	0.004	0.161	2.70	Pass
22	0.001	0.084	0.7	0.001	0.125	0.65	Pass
23	0.003	0.098	2.6	0.003	0.147	1.84	Pass
24	0.000	0.077	0.6	0.001	0.115	0.52	Pass
25	0.005	0.090	5.5	0.005	0.135	3.86	Pass
26	0.001	0.071	1.5	0.001	0.106	1.28	Pass
27	0.005	0.083	6.6	0.006	0.125	4.69	Pass
28	0.001	0.066	1.6	0.001	0.099	1.22	Pass
29	0.005	0.078	6.2	0.005	0.116	4.38	Pass
30	0.001	0.061	2.1	0.002	0.092	1.76	Pass
31	0.003	0.073	4.3	0.003	0.109	3.12	Pass
32	0.001	0.058	1.8	0.001	0.086	1.59	Pass
33	0.003	0.068	4.2	0.003	0.102	3.10	Pass
34	0.001	0.054	1.6	0.001	0.081	1.31	Pass
35	0.002	0.064	3.7	0.003	0.096	2.62	Pass
36	0.001	0.051	1.3	0.001	0.077	1.20	Pass
37	0.003	0.061	4.2	0.003	0.091	2.98	Pass
38	0.001	0.048	1.1	0.001	0.073	1.03	Pass
39	0.001	0.058	2.1	0.001	0.087	1.51	Pass
40	0.000	0.046	1.0	0.001	0.069	0.83	Pass

This test report shall not be reproduced except in full, Without the written approval.

Voltage Source Verification Data (Run time)

EUT: Equipment Under Test Tested by: Test Operator
Test category: Class-A per Ed. 3.0 (2005-11) (European limits) Test Margin: 100
Test date: 2007-10-01 Start time: 오후 9:20:53 End time: 오후 9:23:34
Test duration (min): 2.5 Data file name: H-000279.cts_data
Comment: Comments
Customer: Customer

Test Result: Pass Source qualification: Normal

Highest parameter values during test:

Voltage (Vrms):	229.78	Frequency(Hz):	50.00
I_Peak (Amps):	0.799	I_RMS (Amps):	0.375
I_Fund (Amps):	0.356	Crest Factor:	2.183
Power (Watts):	76.2	Power Factor:	0.885

Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.081	0.460	17.66	OK
3	0.640	2.068	30.97	OK
4	0.080	0.460	17.47	OK
5	0.063	0.919	6.84	OK
6	0.062	0.460	13.44	OK
7	0.058	0.689	8.36	OK
8	0.044	0.460	9.54	OK
9	0.036	0.460	7.84	OK
10	0.032	0.460	6.97	OK
11	0.047	0.230	20.65	OK
12	0.034	0.230	14.96	OK
13	0.040	0.230	17.42	OK
14	0.025	0.230	10.74	OK
15	0.033	0.230	14.43	OK
16	0.034	0.230	14.87	OK
17	0.021	0.230	8.99	OK
18	0.025	0.230	10.77	OK
19	0.026	0.230	11.36	OK
20	0.026	0.230	11.35	OK
21	0.025	0.230	11.00	OK
22	0.013	0.230	5.86	OK
23	0.016	0.230	6.80	OK
24	0.015	0.230	6.49	OK
25	0.015	0.230	6.43	OK
26	0.018	0.230	7.94	OK
27	0.013	0.230	5.77	OK
28	0.013	0.230	5.55	OK
29	0.026	0.230	11.34	OK
30	0.023	0.230	9.99	OK
31	0.014	0.230	6.14	OK
32	0.014	0.230	5.90	OK
33	0.019	0.230	8.43	OK
34	0.020	0.230	8.91	OK
35	0.016	0.230	7.08	OK
36	0.011	0.230	4.97	OK
37	0.014	0.230	5.98	OK
38	0.011	0.230	4.74	OK
39	0.011	0.230	4.94	OK
40	0.011	0.230	4.99	OK

This test report shall not be reproduced except in full, Without the written approval.

6.10 Flicker

6.10.1 Measurement procedure

EUT was connected to the power analyzer system.
Measurement was performed to obtain the desired flicker parameters.
The measuring time depends on which parameters are to be measured.
P_{lt} = 2 h
P_{st} = 10 min
Controls and automatic programs shall be set to produce the most unfavorable sequence of voltage changes, using only those combinations of controls and programs are mentioned by the manufacturer in the instruction manual.

6.10.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Harmonics/Flicker meter	5001x-CTS-400-413	54984	C.I.	08.05.03	☒
Test site	Immunity area	-	-	-	☒

6.10.3 Test data

- Refer to attached test data

Test results

☒ Complied ☐ Not complied

This test report shall not be reproduced except in full, Without the written approval.

- PCI-N804

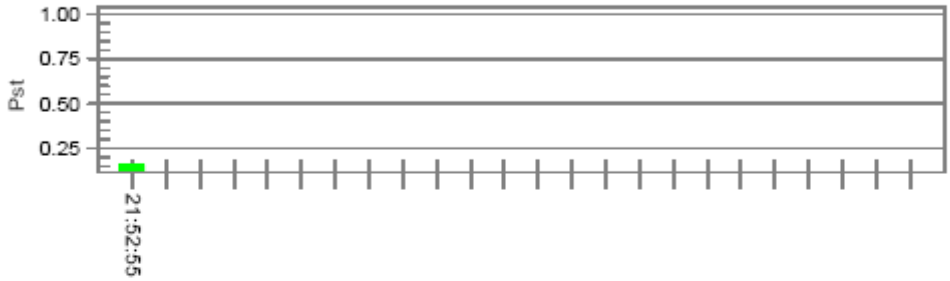
Flicker Test Summary per EN/IEC61000-3-3 (Run time)

EUT: Equipment Under Test
Test category: All parameters (European limits)
Test date: 2007-10-01 Start time: 오후 9:42:35 End time: 오후 9:52:56
Test duration (min): 10 Data file name: F-000283.cts_data
Comment: Comments
Customer: Customer

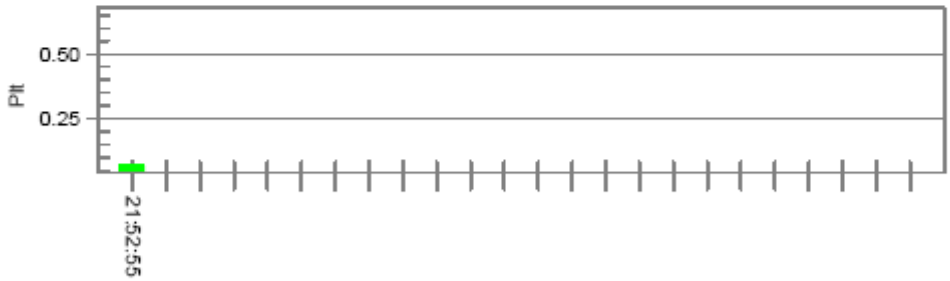
Tested by: Test Operator
Test Margin: 100

Test Result: Pass Status: Test Completed

Pst and limit line European Limits



Plt and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt):	229.55			
Highest dt (%):	0.00	Test limit (%):	3.30	Pass
Time(mS) > dt:	0.0	Test limit (mS):	500.0	Pass
Highest dc (%):	0.00	Test limit (%):	3.30	Pass
Highest dmax (%):	0.00	Test limit (%):	4.00	Pass
Highest Pst (10 min. period):	0.160	Test limit:	1.000	Pass
Highest Plt (2 hr. period):	0.070	Test limit:	0.650	Pass

This test report shall not be reproduced except in full, Without the written approval.

- PCI-N404

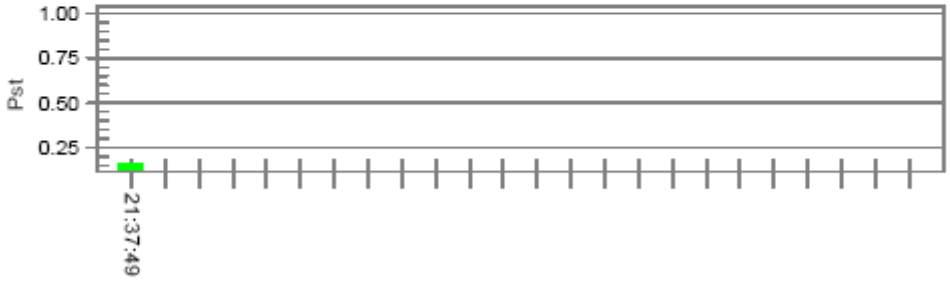
Flicker Test Summary per EN/IEC61000-3-3 (Run time)

EUT: Equipment Under Test
Test category: All parameters (European limits)
Test date: 2007-10-01 Start time: 오후 9:27:30 End time: 오후 9:37:51
Test duration (min): 10 Data file name: F-000281.cts_data
Comment: Comments
Customer: Customer

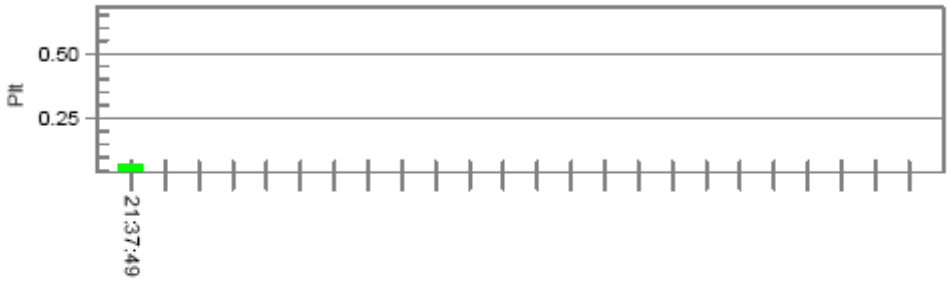
Tested by: Test Operator
Test Margin: 100

Test Result: Pass Status: Test Completed

Pst and limit line European Limits



Plt and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt):	229.55			
Highest dt (%):	0.00	Test limit (%):	3.30	Pass
Time(mS) > dt:	0.0	Test limit (mS):	500.0	Pass
Highest dc (%):	0.00	Test limit (%):	3.30	Pass
Highest dmax (%):	0.00	Test limit (%):	4.00	Pass
Highest Pst (10 min. period):	0.160	Test limit:	1.000	Pass
Highest Plt (2 hr. period):	0.070	Test limit:	0.650	Pass

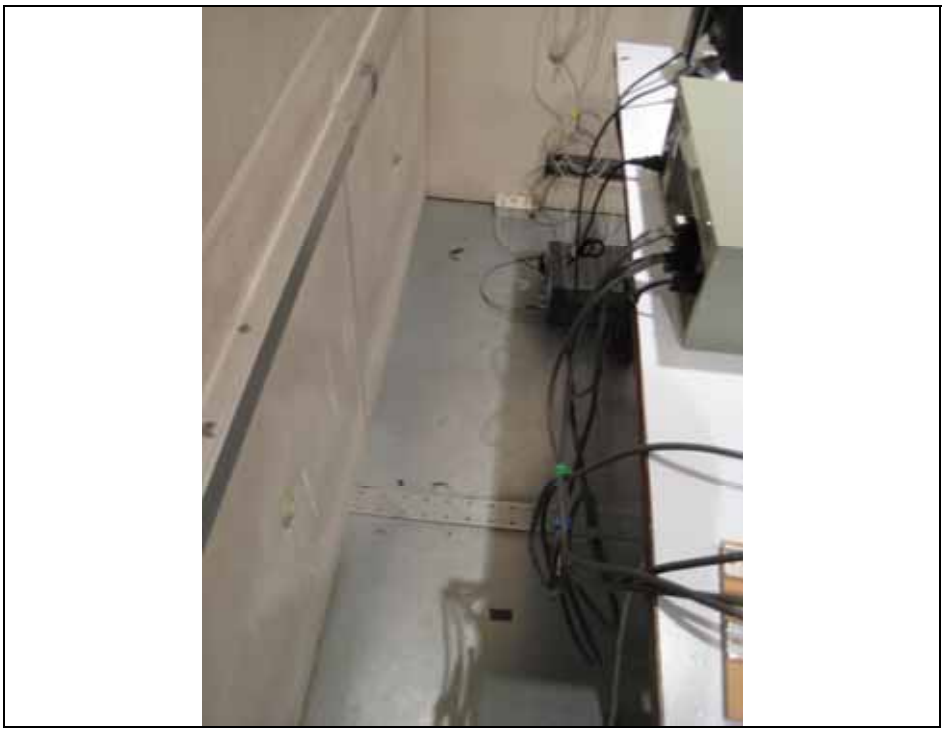
This test report shall not be reproduced except in full, Without the written approval.

7. Test photographs

Conducted emission

- PCI -N804





This test report shall not be reproduced except in full, Without the written approval.

- PCI -N404





This test report shall not be reproduced except in full, Without the written approval.

Radiated Emission

- PCI -N804





This test report shall not be reproduced except in full, Without the written approval.

- PCI -N404





This test report shall not be reproduced except in full, Without the written approval.

Electrostatic Discharge

- PCI -N804



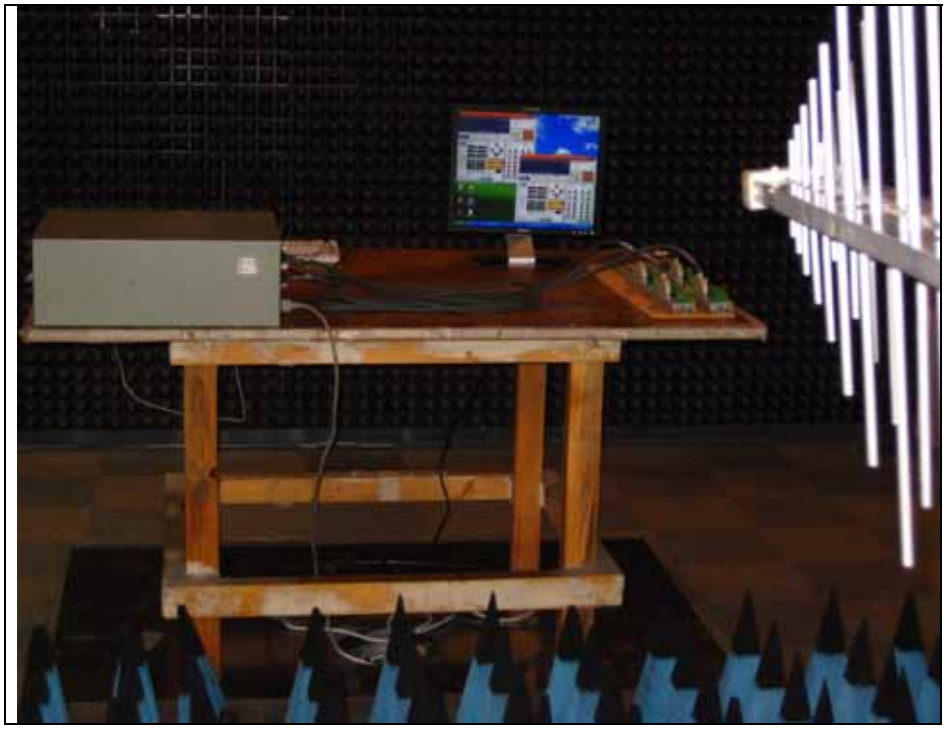
- PCI -N404



This test report shall not be reproduced except in full, Without the written approval.

Radio Frequency Electromagnetic Fields

- PCI -N804



- PCI -N404



This test report shall not be reproduced except in full, Without the written approval.

Electric Fast Transient

- PCI -N804



- PCI -N404



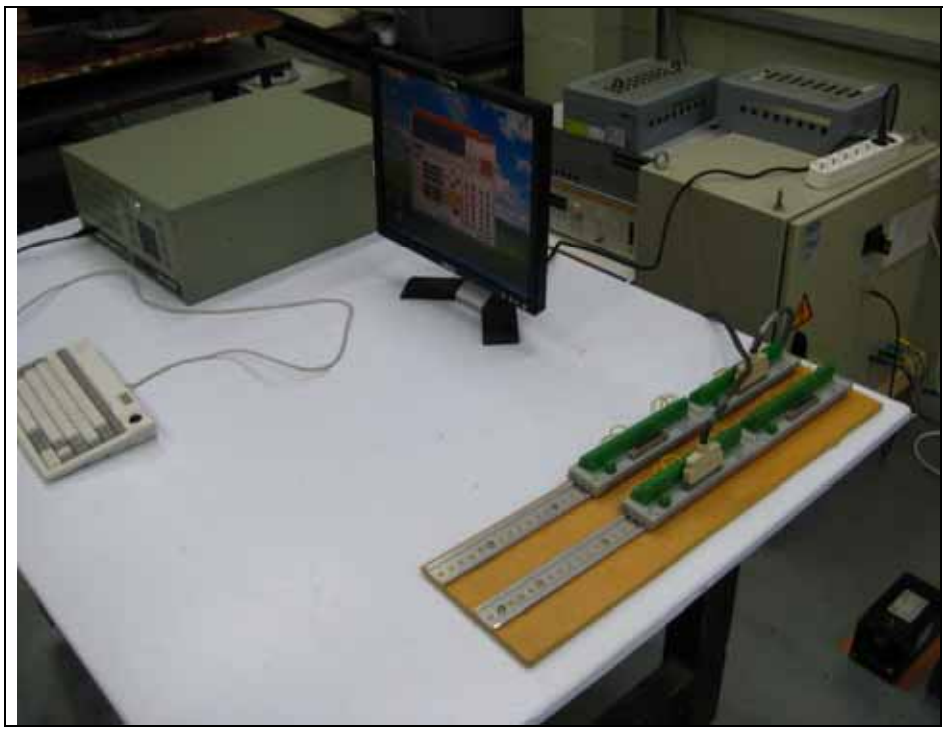
This test report shall not be reproduced except in full, Without the written approval.

Surge

- PCI -N804



- PCI -N404



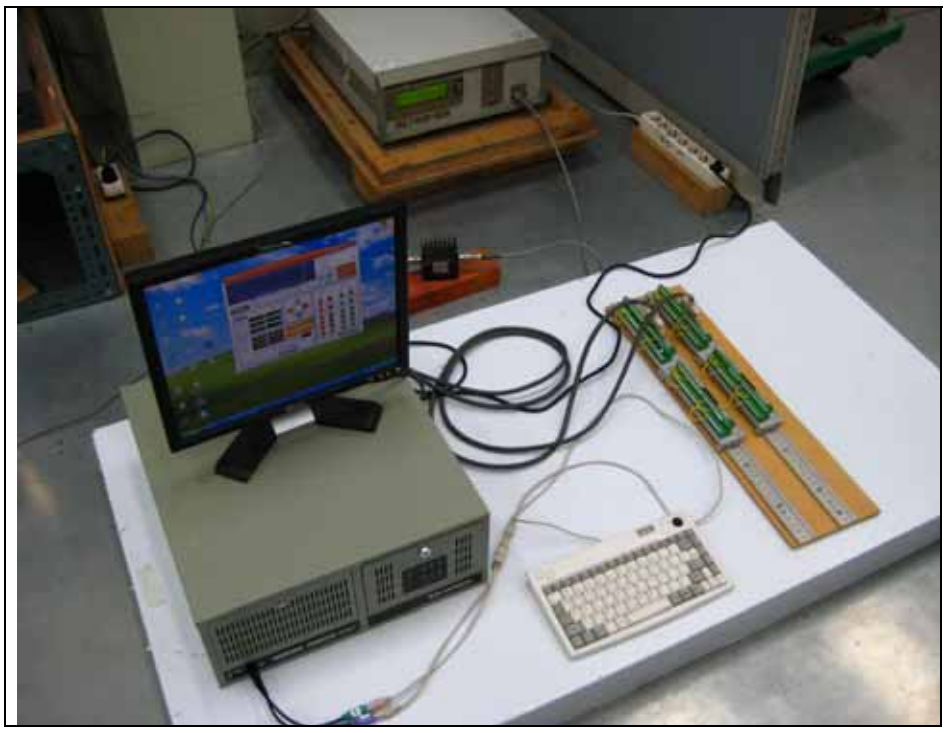
This test report shall not be reproduced except in full, Without the written approval.

Conducted Immunity

- PCI -N804



- PCI -N404



This test report shall not be reproduced except in full, Without the written approval.

Voltage dip/interruption

- PCI -N804



- PCI -N404



This test report shall not be reproduced except in full, Without the written approval.

Harmonics/Fickler

- PCI -N804



- PCI -N404



This test report shall not be reproduced except in full, Without the written approval.

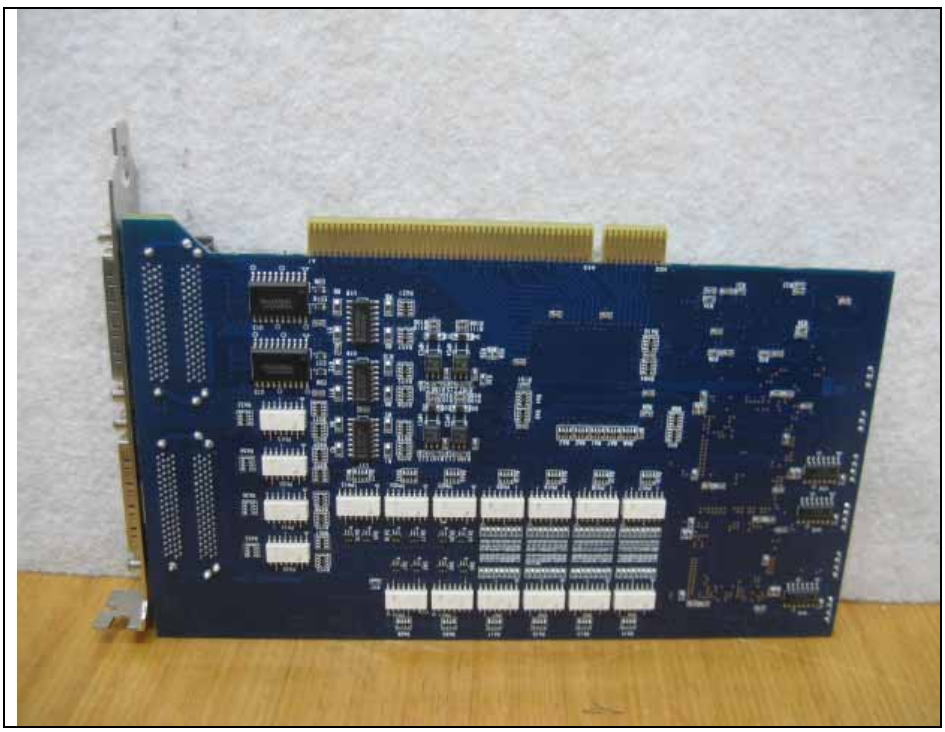
8. E.U.T. photographs

- PCI -N804

Front View



Rear View



This test report shall not be reproduced except in full, Without the written approval.

Inside



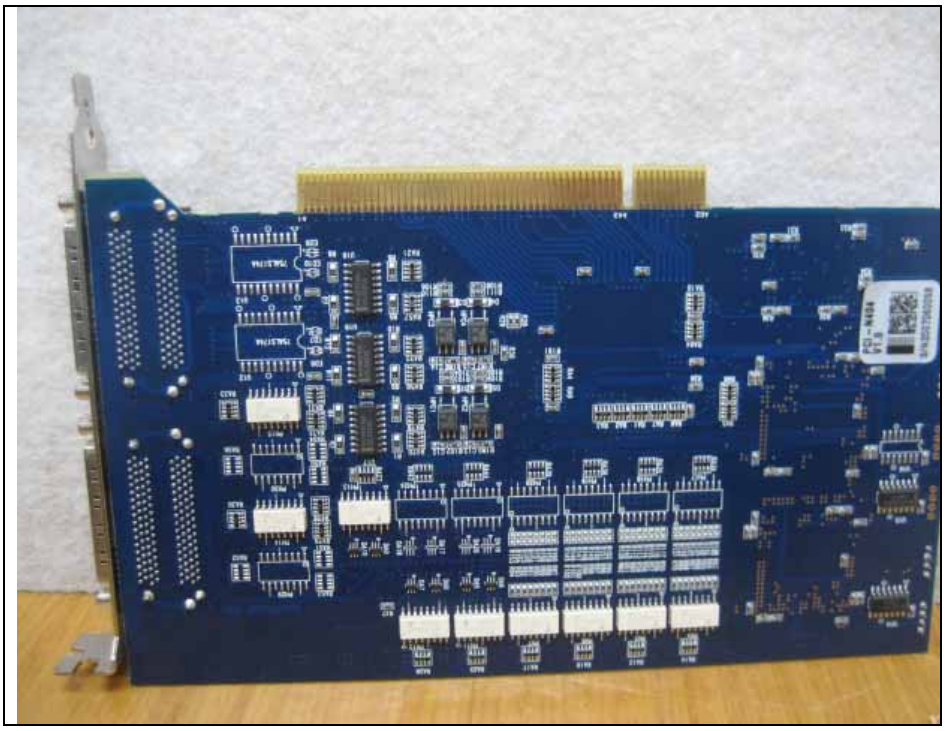
This test report shall not be reproduced except in full, Without the written approval.

- PCI -N404

Front View



Rear View



This test report shall not be reproduced except in full, Without the written approval.

Inside



This test report shall not be reproduced except in full, Without the written approval.